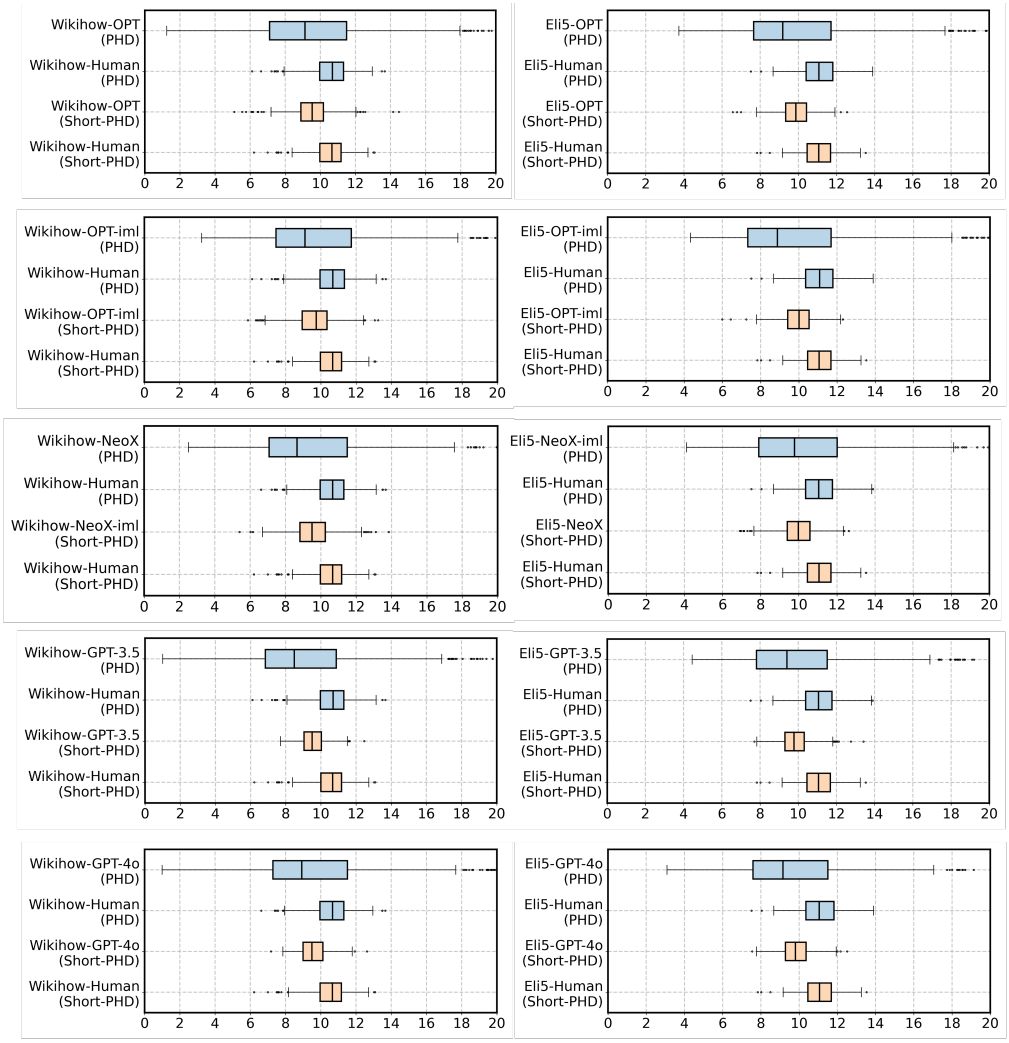

Figure A1: Box plots of PHD distributions by PHD and Short-PHD.

Table A4: Sensitivity Analysis.

(a) Sensitivity Analysis on the Number of Off-topic Content Pieces.

Number	Wikihow	Eli5
1	0.745	0.785
2	0.753	0.812
4	0.759	0.827
6	0.763	0.840
8	0.768	0.844
10	0.771	0.847
12 (default)	0.773	0.849

(b) Sensitivity Analysis on White-box Detection.

Method	Wikihow	Eli5
Roberta	0.730	0.812
Perplexity	0.924	0.959
LogRank	0.897	0.921
DetectGPT	0.911	0.512
DetectLLM	0.478	0.431
PHD	0.602	0.684
Short-PHD	0.760	0.880

Appendix C Related Work

Existing methods for detecting LLM-generated text can be categorized into supervised and zero-shot methods (Wu et al., 2023). Early supervised methods were trained on extracted linguistic or semantic features (Bhatt & Rios, 2021; Sadasivan et al., 2023; Li et al., 2024). Current supervised methods focus on training classifiers to distinguish the neural representations of LLM-generated text from human-written text (Liu et al., 2019; Zellers et al., 2019).

Research has shown that supervised methods can easily overfit to the extracted features or texts in the training dataset (Bakhtin et al., 2019), making them prone to overfitting to the training distribution of specific LLMs and domains. Therefore, we focus on zero-shot detection methods, which require no training and exhibit better generalizability in identifying LLM-generated text across various models and domains (Bao et al., 2024).

Zero-shot methods focus on extracting certain features from large language models and set a threshold to differentiate between LLM-generated and human-written text. For example, one early detection method established a threshold based on the average log probability of a text, which serves as a strong baseline for detecting LLM-generated text (Solaiman et al., 2019). Recent works, such as DetectGPT (Mitchell et al., 2023) and Fast-DetectGPT(Bao et al., 2024), focus on conditional probability curvature, which evaluates the direction of perplexity change after perturbing random tokens. In this work, we investigate a novel detection score called persistent homology dimension (PHD) (Birdal et al., 2021; Tulchinskii et al., 2024), which shows that LLM-generated text tends to have a lower PHD value than human-written text. However, prior studies on PHD face challenges with short texts due to local density peak issues. We address this challenge by proposing a novel method called off-topic content insertion.